Speed of Sound in Air Tube

1. Definition

The speed of sound in air is the distance travelled by a sound wave per unit time through air. It can be determined experimentally using a resonance tube and a tuning fork.

2. Principle

This experiment is based on the principle of resonance. When a vibrating tuning fork is held above an air column, the air column resonates if its natural frequency matches the frequency of the tuning fork, producing a loud sound.

3. Experimental Arrangement

A resonance tube is partially filled with water, and a tuning fork of known frequency is vibrated above its open end. The length of the air column is adjusted by changing the water level until resonance occurs.

For a tube closed at one end by water, the first resonance occurs when the air-column length is approximately one-quarter of the wavelength.

4. Formulae

First resonance:

$L_1 + e = \lambda/4$

Second resonance:

$L_2 + e = 3\lambda/4$

Where:

- L_1 = air-column length at first resonance
- L_2 = air-column length at second resonance
- e = end correction
- λ = wavelength of sound

Subtracting the two equations:

$L_2 - L_1 = \lambda/2$

Therefore,

Wavelength:

$\lambda = 2(L_2 - L_1)$

Speed of sound:

$v = f\lambda$

Hence,

$v = 2f(L_2 - L_1)$

Where:

- v = speed of sound in air

- **f = frequency of the tuning fork**
- **λ = wavelength of sound**

5. Procedure

1. **Fill the resonance tube partially with water.**
2. **Strike the tuning fork gently on a rubber pad.**
3. **Hold the vibrating tuning fork just above the open end of the tube.**
4. **Adjust the water level until the sound becomes loudest.**
5. **Record the first resonance length, L_1.**
6. **Lower the water level further to obtain the second resonance and record L_2.**
7. **Calculate the wavelength and speed of sound using the formulae above.**

6. Precautions

1. **Strike the tuning fork gently using a rubber pad.**
2. **Hold the tuning fork close to the tube opening without touching it.**
3. **Measure the air-column length at eye level.**
4. **Ensure that resonance is identified at the point of maximum loudness.**
5. **Repeat the observations to obtain more accurate results.**

7. Result

The speed of sound in air is calculated using:

$v = 2f(L_2 - L_1)$

At approximately 20°C, the speed of sound in air is about 343 m/s.

8. Conclusion

The speed of sound in air can be determined using a resonance tube by measuring the difference between two successive resonance lengths and multiplying it by twice the tuning fork frequency.